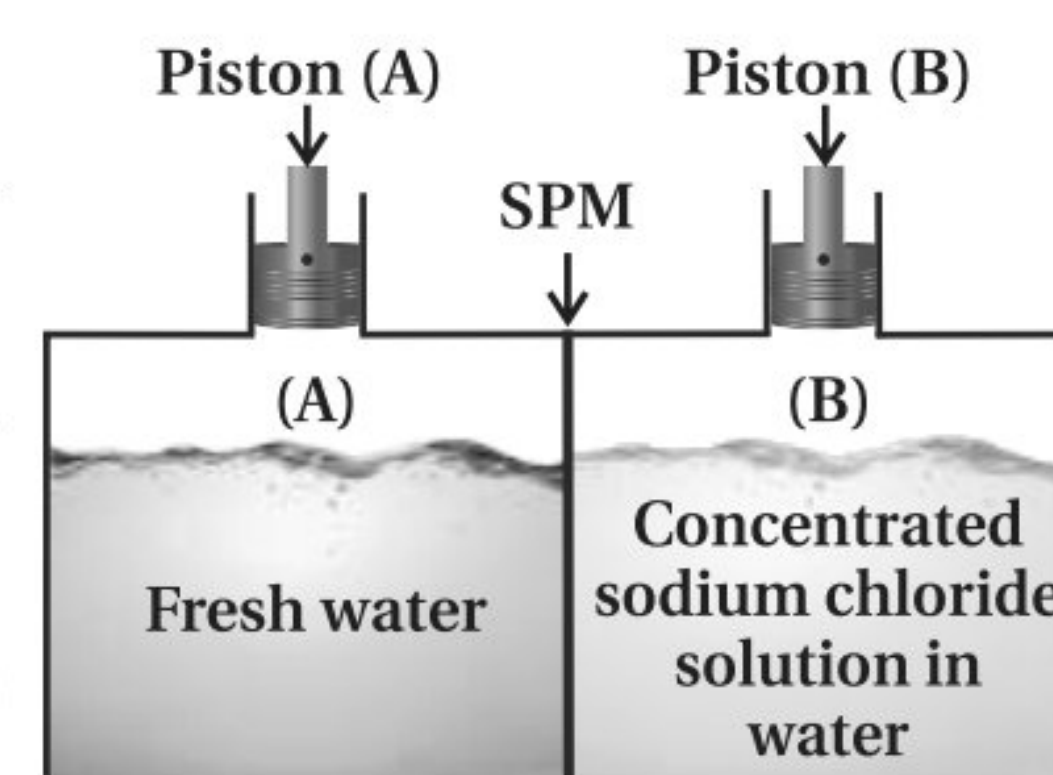


A Choose correct answer from the given options. [Each carries 1 Mark] [40]

- Which of the following aqueous solutions should have the highest boiling point ?
(A) 1.0 M NaOH (B) 1.0 M Na₂SO₄ (C) 1.0 M NH₄NO₃ (D) 1.0 M KNO₃
- The unit of ebullioscopic constant is
(A) K kg mol⁻¹ or K (molality)⁻¹ (B) mol kg K⁻¹ or K⁻¹ (molality)
(C) kg mol⁻¹ K⁻¹ or K⁻¹ (molality)⁻¹ (D) K mol kg⁻¹ or K (molality)
- The values of Van't Hoff factors for KCl, NaCl and K₂SO₄ respectively, are
(A) 2, 2 and 2 (B) 2, 2 and 3 (C) 1, 1 and 2 (D) 1, 1 and 1
- We have three aqueous solutions of NaCl labelled as 'A', 'B' and 'C' with concentrations 0.1 M, 0.01 M and 0.001 M, respectively. The value of Van't Hoff factor for these solutions will be in the order
(A) $i_A < i_B < i_C$ (B) $i_A > i_B > i_C$ (C) $i_A = i_B = i_C$ (D) $i_A < i_B > i_C$
- 4 L of 0.02 M aqueous solution of NaCl was diluted by adding one litre of water. The molarity of the resultant solution is
(A) 0.004 (B) 0.008 (C) 0.012 (D) 0.016
- Consider the Figure and mark the correct option.
(A) Water will move from side (A) to side (B) if a pressure lower than osmotic pressure is applied on piston (B).
(B) Water will move from side (B) to side (A) if a pressure greater than osmotic pressure is applied on piston (B).
(C) Water will move from side (B) to side (A) if a pressure equal to osmotic pressure is applied on piston (B).
(D) Water will move from side (A) to side (B) if pressure equal to osmotic pressure is applied on piston (A).
- Which of the following aqueous solution has same osmotic pressure ?
(i) 0.1 M NaCl solution (ii) 0.1 M glucose solution (iii) 0.6 gm urea in 100 mL solution
(iv) 1.0 gm non-volatile solute (x) in 50 mL solution. (Molecular Mass of x = 200)
(A) (i), (ii), (iii) (B) (ii), (iii), (iv) (C) (i), (ii), (iv) (D) (i), (iii), (iv)
- Osmotic pressure of aqueous solution of urea is 500 mm at 283 K. If its temperature is maintained at 298 K, then on how much dilution its osmotic pressure would be 105.3 mm ?
(A) 10 times (B) 5 times (C) 2.5 times (D) 4 times
- Four solution of K₂SO₄ possess concentration of 0.1 m, 0.01 m, 0.001 m and 0.0001 m respectively. So among which solution has highest van't hoff factor ?
(A) 0.0001 m (B) 0.001 m (C) 0.01 m (D) 0.1 m
- To produce difference between freezing point and boiling point of a solution by 105.0°C, how much sucrose should be dissolved in 100 gm of water ? ($K_f = 1.86^\circ\text{C} \cdot \text{kg mol}^{-1}$ and $K_b = 0.51^\circ\text{C kg mol}^{-1}$)
(A) 72 gm (B) 34.2 gm (C) 342 gm (D) 460 gm
- The total pressure for volatile components A and B is 0.02 bar at equilibrium. If the mole fractions at component A is 0.2, then, what will be the partial pressure of component B ?
(A) 0.02 bar (B) 0.04 bar (C) 0.016 bar (D) 0.2 bar
- 1.5% Urea solution and 5.25% unknown substance's solutions are isotonic. The density of both the solutions are 1 gm/cm³ then what is the molecular weight of unknown substance ?
(A) 90.0 gram/mole (B) 115.0 gram/mole (C) 105.0 gram/mole (D) 210.0 gram/mole



13. How much H_2SO_4 is required to prepare 0.1 M 1 liter H_2SO_4 solution ? The density of 98% W/W H_2SO_4 is 1.80 gm/mL.
(A) 22.20 mL (B) 5.55 mL (C) 11.10 mL (D) 16.65 mL
14. What is the molality of 2.05 molar aqueous solution of acetic acid ? The density of solution is 1.02 gram/mL.
(A) 1.14 M (B) 3.28 M (C) 2.28 M (D) 0.44 M
15. Which of the following having the highest boiling point ?
(A) 0.01 M Na_2SO_4 (B) 0.015 M Sucrose (C) 0.015 M Glucose (D) 0.01 M NaNO_3
16. What will be the osmotic pressure of the solution prepared by dissolving 1.75 gram sucrose in 50 mL solution at 17 °C temperature ?
(A) 0.406 (B) 8.12 (C) 0.822 (D) 0.0406
17. 5% sucrose solution is isotonic with 1% 'X' solution. What will be the molecular weight of 'X' ?
(A) 342 gram/mole (B) 180 gram/mole (C) 68.4 gram/mole (D) 171 gram/mole
18. What is the vapour pressure of the solution prepared by dissolving 0.5 gram substance (Molecular weight 65) in 100 mL CCl_4 ? The vapour pressure of CCl_4 is 143 mm. density of CCl_4 is 1.58.
(A) 143.99 mm (B) 94.39 mm (C) 141.97 mm (D) 199.34 mm
19. The boiling point of water at 760 mm pressure is 373 K. The vapour pressure of water at 298K is 23mm. If the vaporisation enthalpy is 40.656 KJ/mole, then what will be boiling point of water at 23 mm. pressure ?
(A) 250 K (B) 298 K (C) 51.6 K (D) 12.5 K
20. What will be the osmotic pressure of 5 % W/V sugar solution of 150°C temperature ?
(A) 5.07 atmosphere (B) 5.70 atmosphere (C) 7.5 atmosphere (D) 3.55 atmosphere
21. The concentration of an aqueous solution of glucose is 10 % W/V. How much solution is required to dissolve 1 mole glucose ?
(A) 18 litre (B) 9 litre (C) 0.9 litre (D) 1.8 litre
22. What is the mole fraction of NaCl, when solution is prepared by dissolving 5.85 gram NaCl in 90 gram of water ?
(A) 0.2 (B) 0.1 (C) 0.01 (D) 0.0196
23. Which of the following having same Van't Hoff factor as $\text{K}_4[\text{Fe}(\text{CN})_6]$?
(A) $\text{Al}_2(\text{SO}_4)_3$ (B) NaCl (C) $\text{Al}(\text{NO}_3)_3$ (D) Na_2SO_4
24. What will be the Van't Hoff factor i for the 0.1 M ideal solution ?
(A) 0.1 (B) Zero (C) 10 (D) 1
25. Which of the following having same Van't Hoff factor as that of $\text{K}_3[\text{Fe}(\text{CN})_6]$?
(A) $\text{Al}_2(\text{SO}_4)_3$ (B) K_2SO_4 (C) $\text{Al}(\text{NO}_3)_3$ (D) KCl
26. How much methanol should be added to water, to make 150 mL. 2 M solution of CH_3OH ?
(A) 9.6 gram (B) 2.4 gram (C) 9.6×10^3 gram (D) 4.3×10^2 gram
27. Which of the following having maximum boiling point ?
(A) 1.1 M glucose solution (B) 1.1 M Na_2SO_4 solution
(C) 1.1 M KCl solution (D) 1.1 M $\text{C}_6\text{H}_{12}\text{O}_6$ solution
28. What is the concentration of NO_3^- ion on mixing equal volume solution of 0.1 M AgNO_3 and 0.2 M NaCl ?
(A) 0.1 M (B) 0.15 M (C) 0.2 M (D) 0.05 M
29. The solution of 1 gram urea boils at 100.25 °C, At what temperature the same volume solution of 3 gram glucose will boil ?
(A) 100° C (B) 100.5° C (C) 100.25° C (D) 100.75° C
30. The solution of 6.8 gram non-ionic substance in 100 gram water is freezed at -0.93 °C. What is the molar mass of unknown substance ? ($K_f = 1.86$)
(A) 34 (B) 136 (C) 68 (D) 13.6

31. What will be equal for the isotonic solution at constant temperature ?
 (A) Density (B) Normality (C) Molar concentration (D) Volume
32. The molal depression constant of water is 1.86°C/m . Then find out the freezing point of 0.1 m NaCl solution.
 (A) -1.86°C (B) 0.372°C (C) -0.372°C (D) -0.186°C
33. Of the following 0.10 m aqueous solutions, which one will exhibit the largest freezing point depression ?
 (A) KCl (B) $\text{C}_6\text{H}_{12}\text{O}_6$ (C) $\text{Al}_2(\text{SO}_4)_3$ (D) K_2SO_4
34. Which of the following electrolytes has the same value of Van't Hoff's factor (i) as that of $\text{Al}_2(\text{SO}_4)_3$ (if all are 100% ionised) ?
 (A) K_2SO_4 (B) $\text{K}_3[\text{Fe}(\text{CN})_6]$ (C) $\text{Al}(\text{NO}_3)_3$ (D) $\text{K}_4[\text{Fe}(\text{CN})_6]$
35. The boiling point of 0.2 mol kg^{-1} solution of X in water is greater than equimolal solution of Y in water. Which one of the following statements is true in this case ?
 (A) X is undergoing dissociation in water.
 (B) Molecular mass of X is greater than the molecular mass of Y.
 (C) Molecular mass of X is less than the molecular mass of Y.
 (D) Y is undergoing dissociation in water while X undergoes no change.
36. 18 g glucose ($\text{C}_6\text{H}_{12}\text{O}_6$) is added to 178.2 g water. The vapour pressure of water (in torr) for this aqueous solution is :
 (A) 759.0 (B) 7.6 (C) 76.0 (D) 752.4
37. Which one is not equal to zero for an ideal solution ?
 (A) ΔH_{mix} (B) ΔS_{mix}
 (C) ΔV_{mix} (D) $\Delta P = P_{\text{observed}} - P_{\text{Raoult}}$
38. Which of the following statements about the composition of the vapour over an ideal 1 : 1 molar mixture of benzene and toluene is correct ? Assume that the temperature is constant at 25°C . (Given, vapour pressure data at 25°C , benzene = 12.8 kPa toluene = 3.85 kPa)
 (A) The vapour will contain a higher percentage of toluene.
 (B) The vapour will contain equal amounts of benzene and toluene.
 (C) Not enough information is given to make a prediction.
 (D) The vapour will contain a higher percentage of benzene.
39. If increase in boiling point of 1 molal glucose solution is 2K and increase in freezing point of 2 molal glucose solution is also 2K. Then state the relationship of K_b and K_f .
 (A) $K_b = 1.5 K_f$ (B) $K_b = 0.5 K_f$ (C) $K_b = 2K_f$ (D) $K_b = K_f$
40. The mixture that forms maximum boiling azeotrope is :
 (A) Heptane + Octane (B) Water + Nitric acid
 (C) Ethanol + Water (D) Acetone + Carbon disulphide

Brainwave science academy

Std 12 : Chemistry

Unit Test

Chapters : 1

Total Marks : 40

Date : 07/06/26

Time : 03 Hour

Section [A] : 1 Mark MCQ

No	Ans	Chap	Sec	Que	Universal_Queld
1.	B	Chap 1	S4	9	QP26P11B1212_P1C1S4Q9
2.	A	Chap 1	S4	10	QP26P11B1212_P1C1S4Q10
3.	B	Chap 1	S4	15	QP26P11B1212_P1C1S4Q15
4.	A	Chap 1	S4	20	QP26P11B1212_P1C1S4Q20
5.	D	Chap 1	S4	24	QP26P11B1212_P1C1S4Q24
6.	B	Chap 1	S4	19	QP26P11B1212_P1C1S4Q19
7.	B	Chap 1	S8	36	QP26P11B1212_P1C1S8Q36
8.	B	Chap 1	S8	35	QP26P11B1212_P1C1S8Q35
9.	A	Chap 1	S8	34	QP26P11B1212_P1C1S8Q34
10.	A	Chap 1	S8	20	QP26P11B1212_P1C1S8Q20
11.	C	Chap 1	S8	5	QP26P11B1212_P1C1S8Q5
12.	D	Chap 1	S9	31	QP26P11B1212_P1C1S9Q31
13.	B	Chap 1	S9	30	QP26P11B1212_P1C1S9Q30
14.	C	Chap 1	S9	28	QP26P11B1212_P1C1S9Q28
15.	A	Chap 1	S9	25	QP26P11B1212_P1C1S9Q25
16.	C	Chap 1	S9	22	QP26P11B1212_P1C1S9Q22
17.	C	Chap 1	S9	19	QP26P11B1212_P1C1S9Q19
18.	C	Chap 1	S9	17	QP26P11B1212_P1C1S9Q17
19.	B	Chap 1	S9	15	QP26P11B1212_P1C1S9Q15
20.	A	Chap 1	S9	14	QP26P11B1212_P1C1S9Q14
21.	D	Chap 1	S9	12	QP26P11B1212_P1C1S9Q12
22.	D	Chap 1	S9	11	QP26P11B1212_P1C1S9Q11
23.	A	Chap 1	S9	10	QP26P11B1212_P1C1S9Q10
24.	D	Chap 1	S9	9	QP26P11B1212_P1C1S9Q9
25.	C	Chap 1	S9	8	QP26P11B1212_P1C1S9Q8
26.	A	Chap 1	S9	7	QP26P11B1212_P1C1S9Q7
27.	B	Chap 1	S9	6	QP26P11B1212_P1C1S9Q6
28.	D	Chap 1	S9	5	QP26P11B1212_P1C1S9Q5
29.	C	Chap 1	S9	4	QP26P11B1212_P1C1S9Q4
30.	B	Chap 1	S9	3	QP26P11B1212_P1C1S9Q3
31.	C	Chap 1	S9	2	QP26P11B1212_P1C1S9Q2
32.	C	Chap 1	S9	1	QP26P11B1212_P1C1S9Q1

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33.	C	Chap 1	S9	42	QP26P11B1212_P1C1S9Q42
34.	D	Chap 1	S9	43	QP26P11B1212_P1C1S9Q43
35.	A	Chap 1	S9	44	QP26P11B1212_P1C1S9Q44
36.	D	Chap 1	S9	46	QP26P11B1212_P1C1S9Q46
37.	B	Chap 1	S9	45	QP26P11B1212_P1C1S9Q45
38.	D	Chap 1	S9	47	QP26P11B1212_P1C1S9Q47
39.	C	Chap 1	S9	53	QP26P11B1212_P1C1S9Q53
40.	B	Chap 1	S9	54	QP26P11B1212_P1C1S9Q54

Brainwave science academy				
Chapters : 1		Std 12 : Chemistry		Date : 07/06/26
Total Marks : 40		Unit Test		Time : 01 Hour
Student Name : _____				
Roll Number : 012 0123456789 0123456789				
A B C D 1. ☐ ☐ ☐ ☐ 2. ☐ ☐ ☐ ☐ 3. ☐ ☐ ☐ ☐ 4. ☐ ☐ ☐ ☐ 5. ☐ ☐ ☐ ☐ 6. ☐ ☐ ☐ ☐ 7. ☐ ☐ ☐ ☐ 8. ☐ ☐ ☐ ☐ 9. ☐ ☐ ☐ ☐ 10. ☐ ☐ ☐ ☐ <td> A B C D 11. ☐ ☐ ☐ ☐ 12. ☐ ☐ ☐ ☐ 13. ☐ ☐ ☐ ☐ 14. ☐ ☐ ☐ ☐ 15. ☐ ☐ ☐ ☐ 16. ☐ ☐ ☐ ☐ 17. ☐ ☐ ☐ ☐ 18. ☐ ☐ ☐ ☐ 19. ☐ ☐ ☐ ☐ 20. ☐ ☐ ☐ ☐ <td> A B C D 21. ☐ ☐ ☐ ☐ 22. ☐ ☐ ☐ ☐ 23. ☐ ☐ ☐ ☐ 24. ☐ ☐ ☐ ☐ 25. ☐ ☐ ☐ ☐ 26. ☐ ☐ ☐ ☐ 27. ☐ ☐ ☐ ☐ 28. ☐ ☐ ☐ ☐ 29. ☐ ☐ ☐ ☐ 30. ☐ ☐ ☐ ☐ <td> A B C D 31. ☐ ☐ ☐ ☐ 32. ☐ ☐ ☐ ☐ 33. ☐ ☐ ☐ ☐ 34. ☐ ☐ ☐ ☐ 35. ☐ ☐ ☐ ☐ 36. ☐ ☐ ☐ ☐ 37. ☐ ☐ ☐ ☐ 38. ☐ ☐ ☐ ☐ 39. ☐ ☐ ☐ ☐ 40. ☐ ☐ ☐ ☐ <td></td></td></td></td>	A B C D 11. ☐ ☐ ☐ ☐ 12. ☐ ☐ ☐ ☐ 13. ☐ ☐ ☐ ☐ 14. ☐ ☐ ☐ ☐ 15. ☐ ☐ ☐ ☐ 16. ☐ ☐ ☐ ☐ 17. ☐ ☐ ☐ ☐ 18. ☐ ☐ ☐ ☐ 19. ☐ ☐ ☐ ☐ 20. ☐ ☐ ☐ ☐ <td> A B C D 21. ☐ ☐ ☐ ☐ 22. ☐ ☐ ☐ ☐ 23. ☐ ☐ ☐ ☐ 24. ☐ ☐ ☐ ☐ 25. ☐ ☐ ☐ ☐ 26. ☐ ☐ ☐ ☐ 27. ☐ ☐ ☐ ☐ 28. ☐ ☐ ☐ ☐ 29. ☐ ☐ ☐ ☐ 30. ☐ ☐ ☐ ☐ <td> A B C D 31. ☐ ☐ ☐ ☐ 32. ☐ ☐ ☐ ☐ 33. ☐ ☐ ☐ ☐ 34. ☐ ☐ ☐ ☐ 35. ☐ ☐ ☐ ☐ 36. ☐ ☐ ☐ ☐ 37. ☐ ☐ ☐ ☐ 38. ☐ ☐ ☐ ☐ 39. ☐ ☐ ☐ ☐ 40. ☐ ☐ ☐ ☐ <td></td></td></td>	A B C D 21. ☐ ☐ ☐ ☐ 22. ☐ ☐ ☐ ☐ 23. ☐ ☐ ☐ ☐ 24. ☐ ☐ ☐ ☐ 25. ☐ ☐ ☐ ☐ 26. ☐ ☐ ☐ ☐ 27. ☐ ☐ ☐ ☐ 28. ☐ ☐ ☐ ☐ 29. ☐ ☐ ☐ ☐ 30. ☐ ☐ ☐ ☐ <td> A B C D 31. ☐ ☐ ☐ ☐ 32. ☐ ☐ ☐ ☐ 33. ☐ ☐ ☐ ☐ 34. ☐ ☐ ☐ ☐ 35. ☐ ☐ ☐ ☐ 36. ☐ ☐ ☐ ☐ 37. ☐ ☐ ☐ ☐ 38. ☐ ☐ ☐ ☐ 39. ☐ ☐ ☐ ☐ 40. ☐ ☐ ☐ ☐ <td></td></td>	A B C D 31. ☐ ☐ ☐ ☐ 32. ☐ ☐ ☐ ☐ 33. ☐ ☐ ☐ ☐ 34. ☐ ☐ ☐ ☐ 35. ☐ ☐ ☐ ☐ 36. ☐ ☐ ☐ ☐ 37. ☐ ☐ ☐ ☐ 38. ☐ ☐ ☐ ☐ 39. ☐ ☐ ☐ ☐ 40. ☐ ☐ ☐ ☐ <td></td>	

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Brainwave science academy				
Chapters : 1		Std 12 : Chemistry		Date : 07/06/26
Total Marks : 40		Unit Test		Time : 01 Hour
Student Name : _____				
Roll Number : 012 0123456789 0123456789				
A B C D 1. ☐ ☐ ☐ ☐ 2. ☐ ☐ ☐ ☐ 3. ☐ ☐ ☐ ☐ 4. ☐ ☐ ☐ ☐ 5. ☐ ☐ ☐ ☐ 6. ☐ ☐ ☐ ☐ 7. ☐ ☐ ☐ ☐ 8. ☐ ☐ ☐ ☐ 9. ☐ ☐ ☐ ☐ 10. ☐ ☐ ☐ ☐ <td> A B C D 11. ☐ ☐ ☐ ☐ 12. ☐ ☐ ☐ ☐ 13. ☐ ☐ ☐ ☐ 14. ☐ ☐ ☐ ☐ 15. ☐ ☐ ☐ ☐ 16. ☐ ☐ ☐ ☐ 17. ☐ ☐ ☐ ☐ 18. ☐ ☐ ☐ ☐ 19. ☐ ☐ ☐ ☐ 20. ☐ ☐ ☐ ☐ <td> A B C D 21. ☐ ☐ ☐ ☐ 22. ☐ ☐ ☐ ☐ 23. ☐ ☐ ☐ ☐ 24. ☐ ☐ ☐ ☐ 25. ☐ ☐ ☐ ☐ 26. ☐ ☐ ☐ ☐ 27. ☐ ☐ ☐ ☐ 28. ☐ ☐ ☐ ☐ 29. ☐ ☐ ☐ ☐ 30. ☐ ☐ ☐ ☐ <td> A B C D 31. ☐ ☐ ☐ ☐ 32. ☐ ☐ ☐ ☐ 33. ☐ ☐ ☐ ☐ 34. ☐ ☐ ☐ ☐ 35. ☐ ☐ ☐ ☐ 36. ☐ ☐ ☐ ☐ 37. ☐ ☐ ☐ ☐ 38. ☐ ☐ ☐ ☐ 39. ☐ ☐ ☐ ☐ 40. ☐ ☐ ☐ ☐ <td></td></td></td></td>	A B C D 11. ☐ ☐ ☐ ☐ 12. ☐ ☐ ☐ ☐ 13. ☐ ☐ ☐ ☐ 14. ☐ ☐ ☐ ☐ 15. ☐ ☐ ☐ ☐ 16. ☐ ☐ ☐ ☐ 17. ☐ ☐ ☐ ☐ 18. ☐ ☐ ☐ ☐ 19. ☐ ☐ ☐ ☐ 20. ☐ ☐ ☐ ☐ <td> A B C D 21. ☐ ☐ ☐ ☐ 22. ☐ ☐ ☐ ☐ 23. ☐ ☐ ☐ ☐ 24. ☐ ☐ ☐ ☐ 25. ☐ ☐ ☐ ☐ 26. ☐ ☐ ☐ ☐ 27. ☐ ☐ ☐ ☐ 28. ☐ ☐ ☐ ☐ 29. ☐ ☐ ☐ ☐ 30. ☐ ☐ ☐ ☐ <td> A B C D 31. ☐ ☐ ☐ ☐ 32. ☐ ☐ ☐ ☐ 33. ☐ ☐ ☐ ☐ 34. ☐ ☐ ☐ ☐ 35. ☐ ☐ ☐ ☐ 36. ☐ ☐ ☐ ☐ 37. ☐ ☐ ☐ ☐ 38. ☐ ☐ ☐ ☐ 39. ☐ ☐ ☐ ☐ 40. ☐ ☐ ☐ ☐ <td></td></td></td>	A B C D 21. ☐ ☐ ☐ ☐ 22. ☐ ☐ ☐ ☐ 23. ☐ ☐ ☐ ☐ 24. ☐ ☐ ☐ ☐ 25. ☐ ☐ ☐ ☐ 26. ☐ ☐ ☐ ☐ 27. ☐ ☐ ☐ ☐ 28. ☐ ☐ ☐ ☐ 29. ☐ ☐ ☐ ☐ 30. ☐ ☐ ☐ ☐ <td> A B C D 31. ☐ ☐ ☐ ☐ 32. ☐ ☐ ☐ ☐ 33. ☐ ☐ ☐ ☐ 34. ☐ ☐ ☐ ☐ 35. ☐ ☐ ☐ ☐ 36. ☐ ☐ ☐ ☐ 37. ☐ ☐ ☐ ☐ 38. ☐ ☐ ☐ ☐ 39. ☐ ☐ ☐ ☐ 40. ☐ ☐ ☐ ☐ <td></td></td>	A B C D 31. ☐ ☐ ☐ ☐ 32. ☐ ☐ ☐ ☐ 33. ☐ ☐ ☐ ☐ 34. ☐ ☐ ☐ ☐ 35. ☐ ☐ ☐ ☐ 36. ☐ ☐ ☐ ☐ 37. ☐ ☐ ☐ ☐ 38. ☐ ☐ ☐ ☐ 39. ☐ ☐ ☐ ☐ 40. ☐ ☐ ☐ ☐ <td></td>	

A Choose correct answer from the given options. [Each carries 1 Mark] [40]

1. Which of the following aqueous solutions should have the highest boiling point ?

(A) 1.0 M NaOH (B) 1.0 M Na₂SO₄ (C) 1.0 M NH₄NO₃ (D) 1.0 M KNO₃

Ans. (B)

➡ Explanation : In 1.0 M Na₂SO₄ solution Vant Hoff factor, $i > 1$ and is maximum when compared with the other three given electrolytes. So, the extent of dissociation in case of 1.0 M Na₂SO₄ would be highest yielding maximum no. of ions as compared to the other given electrolytes in their 1.0 M solutions.

2. The unit of ebullioscopic constant is

(A) K kg mol⁻¹ or K (molality)⁻¹ (B) mol kg K⁻¹ or K⁻¹ (molality)
(C) kg mol⁻¹ K⁻¹ or K⁻¹ (molality)⁻¹ (D) K mol kg⁻¹ or K (molality)

Ans. (A)

➡ $\Delta T_b = K_b \cdot m$

$$K_b = \frac{\Delta T_b}{m} = \frac{K}{\text{Molality}} = \frac{K}{\text{mol kg}^{-1}} = K \text{ kg mol}^{-1}$$

3. The values of Van't Hoff factors for KCl, NaCl and K₂SO₄ respectively, are

(A) 2, 2 and 2 (B) 2, 2 and 3 (C) 1, 1 and 2 (D) 1, 1 and 1

Ans. (B)

➡ Explanation : The values of Van't Hoff factor depend upon the extent of dissociation. KCl, NaCl and K₂SO₄ are strong electrolytes. The extent or degree of dissociation would be greatest with Na₂SO₄ as compared to that of KCl and NaCl.

4. We have three aqueous solutions of NaCl labelled as 'A', 'B' and 'C' with concentrations 0.1 M, 0.01 M and 0.001 M, respectively. The value of Van't Hoff factor for these solutions will be in the order

(A) $i_A < i_B < i_C$ (B) $i_A > i_B > i_C$ (C) $i_A = i_B = i_C$ (D) $i_A < i_B > i_C$

Ans. (A)

➡ Explanation : It is because the extent or degree of dissociation increases with increase in dilution of a solution. 0.001 M NaCl solution 'C' is most diluted as compared to the other two NaCl solutions. The Vant Hoff factor (A) depends on extent of dilution. This confirms answer at (A) $i_A < i_B < i_C$

5. 4 L of 0.02 M aqueous solution of NaCl was diluted by adding one litre of water. The molarity of the resultant solution is

(A) 0.004 (B) 0.008 (C) 0.012 (D) 0.016

Ans. (D)

➡ Explanation :

Apply the relation : $M_1 V_1 = M_2 V_2$

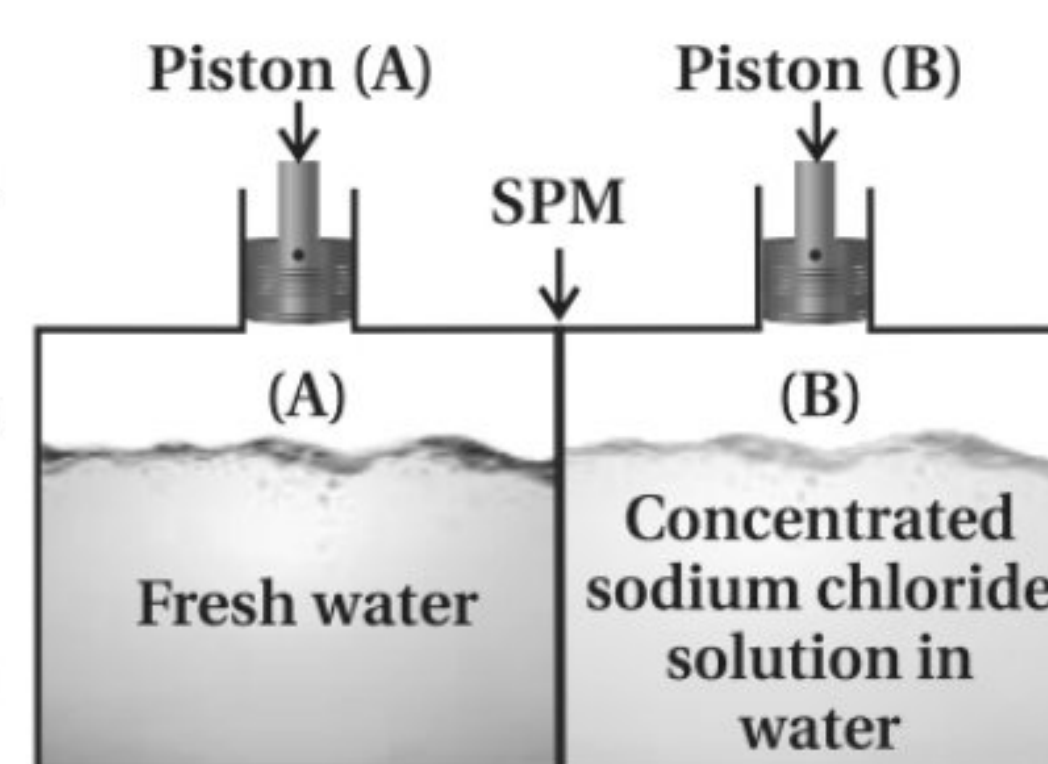
Given : $M_1 = 0.02 \text{ M}$, $V_1 = 4 \text{ L}$, $V_2 = 5 \text{ L}$, $M_2 = ?$

Therefore, $0.02 \times 4 \text{ L} = M_2 \times 5 \text{ L}$

$$M_2 = 0.08/5 = 0.016 \text{ M}$$

6. Consider the Figure and mark the correct option.

- (A) Water will move from side (A) to side (B) if a pressure lower than osmotic pressure is applied on piston (B).
 (B) Water will move from side (B) to side (A) if a pressure greater than osmotic pressure is applied on piston (B).
 (C) Water will move from side (B) to side (A) if a pressure equal to osmotic pressure is applied on piston (B).
 (D) Water will move from side (A) to side (B) if pressure equal to osmotic pressure is applied on piston (A).



Ans. (B)

7. Which of the following aqueous solution has same osmotic pressure ?

- (i) 0.1 M NaCl solution (ii) 0.1 M glucose solution (iii) 0.6 gm urea in 100 mL solution
 (iv) 1.0 gm non-volatile solute (x) in 50 mL solution. (Molecular Mass of x = 200)
 (A) (i), (ii), (iii) (B) (ii), (iii), (iv) (C) (i), (ii), (iv) (D) (i), (iii), (iv)

Ans. (B) (ii), (iii), (iv)

⇒ (ii) 0.1 M glucose = $\pi = 0.1 RT$

$$(iii) \pi = \frac{n}{V} RT = \frac{\frac{0.6}{60} \times 1000}{100} RT = 0.1 RT$$

$$(iv) \pi = \frac{0.1}{\frac{200}{50}} \times 1000 RT = 0.1 RT$$

8. Osmotic pressure of aqueous solution of urea is 500 mm at 283 K. If its temperature is maintained at 298 K, then on how much dilution its osmotic pressure would be 105.3 mm ?

- (A) 10 times (B) 5 times (C) 2.5 times (D) 4 times

Ans. (B) 5 times

$$\Rightarrow \frac{\pi_1}{\pi_2} = \frac{T_1}{T_2} \times \frac{V_2}{V_1}$$

$$\therefore \frac{500/760}{105.3/760} = \frac{283 \times V_2}{V_1 \times 298}$$

$$\therefore \frac{V_2}{V_1} = 5$$

9. Four solution of K_2SO_4 possess concentration of 0.1 m, 0.01 m, 0.001 m and 0.0001 m respectively. So among which solution has highest van't Hoff factor ?

- (A) 0.0001 m (B) 0.001 m (C) 0.01 m (D) 0.1 m

Ans. (A) 0.0001 m

10. To produce difference between freezing point and boiling point of a solution by 105.0°C , how much sucrose should be dissolved in 100 gm of water ? ($K_f = 1.86^\circ\text{C} \cdot \text{kg mol}^{-1}$ and $K_b = 0.51^\circ\text{C} \cdot \text{kg mol}^{-1}$)

- (A) 72 gm (B) 34.2 gm (C) 342 gm (D) 460 gm

Ans. (A) 72 gm

$$\Rightarrow \text{Boiling point } (T_b) = 100 + \Delta T_b \\ = 100 + K_b \cdot m$$

$$\Rightarrow \text{Freezing point } (T_f) = 0 - \Delta T_f = -K_f \cdot m$$

$$\therefore T_b - T_f = (100 + K_b \cdot m) - (-K_f \cdot m)$$

$$105 = 100 + 0.51 m + 1.86 m$$

$$2.37 m = 5$$

$$m = \frac{5}{2.37} = 2.11$$

➡ So, mass of sucrose to dissolve in 100 gm water is, $= \frac{2.11 \times 342 \times 100}{1000} = 72 \text{ gm}$

11. The total pressure for volatile components A and B is 0.02 bar at equilibrium. If the mole fractions at component A is 0.2, then, what will be the partial pressure of component B ?

(A) 0.02 bar (B) 0.04 bar (C) 0.016 bar (D) 0.2 bar

Ans. (C) 0.016 bar

$$\Rightarrow X_A + X_B = 1$$

$$\therefore 0.2 + X_B = 1 \quad \therefore X_B = 0.8$$

$$p_B = p \cdot X_B = (0.02) (0.8)$$

$$\therefore p_B = 0.016 \text{ bar}$$

12. 1.5% Urea solution and 5.25% unknown substance's solutions are isotonic. The density of both the solutions are 1 gm/cm^3 then what is the molecular weight of unknown substance ?

(A) 90.0 gram/mole (B) 115.0 gram/mole (C) 105.0 gram/mole (D) 210.0 gram/mole

Ans. (D) 210.0 gram/mole

$$\Rightarrow \text{For isotonic solution } \frac{W_{\text{unknown}}}{M_{\text{unknown}}} = \frac{W_{\text{urea}}}{M_{\text{urea}}}$$

$$M_{\text{unknown}} = \frac{5.25 \times 60}{1.5} = 210 \text{ gram/mole}$$

13. How much H_2SO_4 is required to prepare 0.1 M 1 liter H_2SO_4 solution ? The density of 98% W/W H_2SO_4 is 1.80 gm/mL .

(A) 22.20 mL (B) 5.55 mL (C) 11.10 mL (D) 16.65 mL

Ans. (B) 5.55 mL

$$\Rightarrow M = \frac{\% \text{ of solute} \times 10 \times \text{density}}{\text{molecular weight}}$$

$$M = \frac{98 \times 1.80 \times 10}{98} = 18 \text{ M}$$

$$M_1 V_1 = M_2 V_2$$

$$18 \times V_1 = 0.1 \times 1000$$

$$V_1 = \frac{0.1 \times 1000}{18} = 5.55 \text{ mL}$$

14. What is the molality of 2.05 molar aqueous solution of acetic acid ? The density of solution is 1.02 gram/mL .

(A) 1.14 M (B) 3.28 M (C) 2.28 M (D) 0.44 M

Ans. (C) 2.28 m

$$\Rightarrow M = 2.05 \text{ M}, \quad d = 1.02, \quad M_1 = 60 \text{ g/mol}$$

$$m = \frac{1000 M}{1000d - MM_1} = \frac{1000 \times 2.05}{1000 \times 1.02 - (2.05 \times 60)} = 2.28 \text{ m}$$

15. Which of the following having the highest boiling point ?

(A) 0.01 M Na_2SO_4 (B) 0.015 M Sucrose (C) 0.015 M Glucose (D) 0.01 M NaNO_3

Ans. (A) 0.01 M Na_2SO_4

16. What will be the osmotic pressure of the solution prepared by dissolving 1.75 gram sucrose in 50 mL solution at 17°C temperature ?

(A) 0.406 (B) 8.12 (C) 0.822 (D) 0.0406

Ans. (C) 0.822

→ $T = 17^{\circ}\text{C} = 293 \text{ K}, w = 1.75\text{g}, V = 0.15 \text{ l}$

$$\pi = \frac{WRT}{MV}$$

$$= \frac{1.75 \times 0.08314 \times 293}{342 \times 0.15}$$

$$= 0.822 \text{ bar}$$

17. 5% sucrose solution is isotonic with 1% 'X' solution. What will be the molecular weight of 'X' ?
 (A) 342 gram/mole (B) 180 gram/mole (C) 68.4 gram/mole (D) 171 gram/mole

Ans. (C) 68.4 gram/mole

→ For the 5% solution $\frac{5}{M_1}$ and 1% solution = $\frac{1}{M_2}$

$$\therefore \frac{5}{342} = \frac{1}{M_2}$$

$$\therefore M_2 = 68.4 \text{ gram/mole}$$

18. What is the vapour pressure of the solution prepared by dissolving 0.5 gram substance (Molecular weight 65) in 100 mL CCl_4 ? The vapour pressure of CCl_4 is 143 mm. density of CCl_4 is 1.58.
 (A) 143.99 mm (B) 94.39 mm (C) 141.97 mm (D) 199.34 mm

Ans. (C) 141.97 mm

→ $w_B = 0.5 \text{ gram}$
 $w_A = \text{volume} \times \text{density}$
 $= 100 \times 1.58 = 158 \text{ gram } \text{CCl}_4$
 $M_B = 65 \text{ gram/volume}$
 $M_A = 154 \text{ gram/volume}$
 $\frac{p_1^0 - p_1}{p^0} = \frac{w_B \cdot M_A}{w_A \cdot M_B} = \frac{0.5 \times 154}{158 \times 65}$
 $\therefore p = 141.97 \text{ mm}$

19. The boiling point of water at 760 mm pressure is 373 K. The vapour pressure of water at 298K is 23mm. If the vaporisation enthalpy is 40.656 KJ/mole, then what will be boiling point of water at 23 mm. pressure ?
 (A) 250 K (B) 298 K (C) 51.6 K (D) 12.5 K

Ans. (B) 298 K

→ Boiling point is the temperature at which the vapour pressure will be equal to the atmospheric pressure.

20. What will be the osmotic pressure of 5 % W/V sugar solution of 150°C temperature ?
 (A) 5.07 atmosphere (B) 5.70 atmosphere (C) 7.5 atmosphere (D) 3.55 atmosphere

Ans. (A) 5.07 atmosphere

→ $\pi = \frac{5 \times 0.082 \times 423}{342 \times 0.1} = 5.07 \text{ atm.}$

21. The concentration of an aqueous solution of glucose is 10 % W/V. How much solution is required to dissolve 1 mole glucose ?
 (A) 18 litre (B) 9 litre (C) 0.9 litre (D) 1.8 litre

Ans. (D) 1.8 litre

→ $10\% \text{ W/V} = \frac{180 \times 100}{x}$
 $x = 1800 \text{ mL} = 1.8 \text{ litre}$

22. What is the mole fraction of NaCl, when solution is prepared by dissolving 5.85 gram NaCl in 90 gram of water ?

(A) 0.2

(B) 0.1

(C) 0.01

(D) 0.0196

Ans. (D) 0.0196

$$\Rightarrow X_{\text{NaCl}} = \frac{5.85/58.5}{\frac{90}{18} + \frac{5.85}{58.5}} = \frac{0.1}{5.1} = 0.0196$$

23. Which of the following having same Van't Hoff factor as $\text{K}_4[\text{Fe}(\text{CN})_6]$?(A) $\text{Al}_2(\text{SO}_4)_3$ (B) NaCl (C) $\text{Al}(\text{NO}_3)_3$ (D) Na_2SO_4 Ans. (A) $\text{Al}_2(\text{SO}_4)_3$ 24. What will be the Van't Hoff factor i for the 0.1 M ideal solution ?

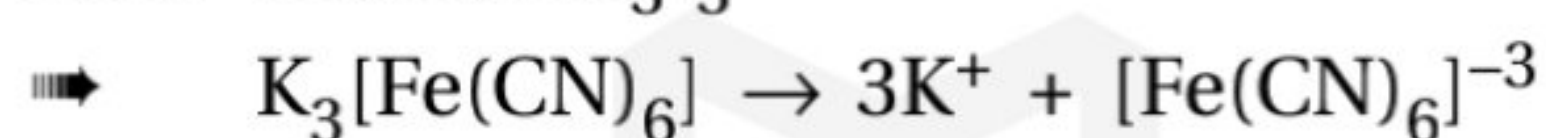
(A) 0.1

(B) Zero

(C) 10

(D) 1

Ans. (D) 1

25. Which of the following having same Van't Hoff factor as that of $\text{K}_3[\text{Fe}(\text{CN})_6]$?(A) $\text{Al}_2(\text{SO}_4)_3$ (B) K_2SO_4 (C) $\text{Al}(\text{NO}_3)_3$ (D) KCl Ans. (C) $\text{Al}(\text{NO}_3)_3$ 

$$\Rightarrow \text{Total no. of ions in both are same.}$$
26. How much methanol should be added to water, to make 150 mL 2 M solution of CH_3OH ?

(A) 9.6 gram

(B) 2.4 gram

(C) 9.6×10^3 gram(D) 4.3×10^2 gram

Ans. (A) 9.6 gram



$$= \frac{32 \times 150 \times 2}{1000} = 9.6 \text{ gram}$$

27. Which of the following having maximum boiling point ?

(A) 1.1 M glucose solution

(B) 1.1 M Na_2SO_4 solution(C) 1.1 M KCl solution(D) 1.1 M $\text{C}_6\text{H}_{12}\text{O}_6$ solutionAns. (B) 1.1 M Na_2SO_4 solution28. What is the concentration of NO_3^- ion on mixing equal volume solution of 0.1 M AgNO_3 and 0.2 M NaCl ?

(A) 0.1 M

(B) 0.15 M

(C) 0.2 M

(D) 0.05 M

Ans. (D) 0.05 M



So total volume = 2 litre

So concentration will be halved.

$$\therefore \text{Concentration of } \text{AgNO}_3 = \frac{0.1}{2} = 0.05 \text{ M}$$

29. The solution of 1 gram urea boils at 100.25°C , At what temperature the same volume solution of 3 gram glucose will boil ?(A) 100°C (B) 100.5°C (C) 100.25°C (D) 100.75°C Ans. (C) 100.25°C

$$\Rightarrow \text{Mole of urea} = \frac{1}{60}$$

$$\text{Mole of glucose} = \frac{3}{180} = \frac{1}{60}$$

$$\therefore \text{Concentration is same and hence boiling point also same.}$$
30. The solution of 6.8 gram non-ionic substance in 100 gram water is freezed at -0.93°C . What is the molar mass of unknown substance ? ($K_f = 1.86$)

(A) 34

(B) 136

(C) 68

(D) 13.6

Ans. (B) 136

$$\begin{aligned}
 \Rightarrow M &= \frac{K_f \times w \times 1000}{\Delta T_f \times w_A} \\
 &= \frac{1.86 \times 6.8 \times 1000}{0.93 \times 100} \\
 &= 136 \text{ gram/mole}
 \end{aligned}$$

31. What will be equal for the isotonic solution at constant temperature ?

(A) Density

(B) Normality

(C) Molar concentration

(D) Volume

Ans. (C) Molar concentration

32. The molal depression constant of water is 1.86°C/m . Then find out the freezing point of 0.1 m NaCl solution.(A) -1.86°C (B) 0.372°C (C) -0.372°C (D) -0.186°C Ans. (C) -0.372°C $\Rightarrow$ For NaCl, $i = 2$

$$\begin{aligned}
 \Delta T_f &= 2K_f \cdot m \\
 &= 2 \times 1.86 \times 0.1 = 0.372
 \end{aligned}$$

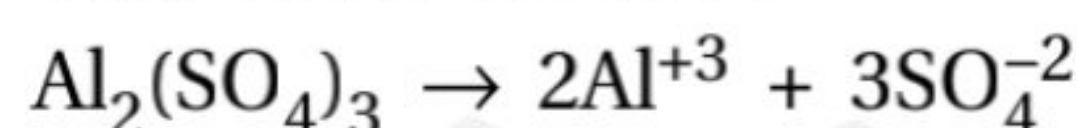
$$\begin{aligned}
 \Rightarrow T_s &= T - \Delta T_f \\
 &= 0 - 0.372 = -0.372^\circ \text{C}
 \end{aligned}$$

33. Of the following 0.10 m aqueous solutions, which one will exhibit the largest freezing point depression ?

(A) KCl

(B) $\text{C}_6\text{H}_{12}\text{O}_6$ (C) $\text{Al}_2(\text{SO}_4)_3$ (D) K_2SO_4 Ans. (C) $\text{Al}_2(\text{SO}_4)_3$

$$\Rightarrow \Delta T_f = iK_fm$$

 i is highest for $\text{Al}_2(\text{SO}_4)_3$ 34. Which of the following electrolytes has the same value of Van't Hoff's factor (i) as that of $\text{Al}_2(\text{SO}_4)_3$ (if all are 100% ionised) ?(A) K_2SO_4 (B) $\text{K}_3[\text{Fe}(\text{CN})_6]$ (C) $\text{Al}(\text{NO}_3)_3$ (D) $\text{K}_4[\text{Fe}(\text{CN})_6]$ Ans. (D) $\text{K}_4[\text{Fe}(\text{CN})_6]$ $\Rightarrow$ Van't Hoff factor ofSo, $n = 5$ So, $n = 5 \Rightarrow i = n = 5$ 35. The boiling point of 0.2 mol kg^{-1} solution of X in water is greater than equimolal solution of Y in water. Which one of the following statements is true in this case ?

(A) X is undergoing dissociation in water.

(B) Molecular mass of X is greater than the molecular mass of Y.

(C) Molecular mass of X is less than the molecular mass of Y.

(D) Y is undergoing dissociation in water while X undergoes no change.

Ans. (A) X is undergoing dissociation in water.

 $\Rightarrow (\Delta T_b)_x > (\Delta T_b)_y$ same solvent so, K_b is same. m is same (given).

$$i_x \cdot K_b \cdot m > i_y \cdot K_b \cdot m \Rightarrow i_x > i_y$$

 $\Rightarrow$ So, X is undergoing dissociation in water.36. 18 g glucose ($\text{C}_6\text{H}_{12}\text{O}_6$) is added to 178.2 g water. The vapour pressure of water (in torr) for this aqueous solution is :

(A) 759.0

(B) 7.6

(C) 76.0

(D) 752.4

Ans. (D) 752.4

➡ From Rault's law,

$$\frac{p_1^0 - p_1}{p_1^0} = X_{\text{solute}}$$

$$\therefore \frac{p_1^0 - p_1}{p_1^0} = \frac{n_{\text{solute}}}{n_{\text{solute}} + N_{\text{solvent}}}$$

$$\therefore \frac{760 - p_1}{760} = \frac{0.1}{\frac{178.2}{18} + 0.1} = 0.01$$

$$(\because p_0 = 760 \text{ torr } (100^\circ\text{C}))$$

$$\therefore 760 - p_1 = 7.6$$

$$\therefore p_1 = 752.4$$

$$\therefore \text{Vapour pressure of H}_2\text{O in solution} = 752.3 \text{ torr}$$

37. Which one is not equal to zero for an ideal solution ?

(A) ΔH_{mix}

(B) ΔS_{mix}

(C) ΔV_{mix}

(D) $\Delta P = P_{\text{observed}} - P_{\text{Raoult}}$

Ans. (B) ΔS_{mix}

➡ For an ideal solution $\Delta S_{\text{mix}} > 0$

38. Which of the following statements about the composition of the vapour over an ideal 1 : 1 molar mixture of benzene and toluene is correct ? Assume that the temperature is constant at 25°C . (Given, vapour pressure data at 25°C , benzene = 12.8 kPa toluene = 3.85 kPa)

(A) The vapour will contain a higher percentage of toluene.

(B) The vapour will contain equal amounts of benzene and toluene.

(C) Not enough information is given to make a prediction.

(D) The vapour will contain a higher percentage of benzene.

Ans. (D) The vapour will contain a higher percentage of benzene.

➡ In comparison vapour pressure of benzene is more hence its proportion of percentage is more in mixture.

39. If increase in boiling point of 1 molal glucose solution is 2K and increase in freezing point of 2 molal glucose solution is also 2K. Then state the relationship of K_b and K_f .

(A) $K_b = 1.5 K_f$

(B) $K_b = 0.5 K_f$

(C) $K_b = 2K_f$

(D) $K_b = K_f$

Ans. (C) $K_b = 2K_f$

$$\frac{\Delta T_b}{\Delta T_f} = \frac{i \times m \times K_b}{i \times m \times K_f}$$

$$\therefore \frac{2}{2} = \frac{1 \times 1 \times K_b}{1 \times 2 \times K_f} \quad \therefore K_b = 2K_f$$

40. The mixture that forms maximum boiling azeotrope is :

(A) Heptane + Octane

(B) Water + Nitric acid

(C) Ethanol + Water

(D) Acetone + Carbon disulphide

Ans. (B) Water + Nitric acid

➡ Negative deviated mixtures from ideal behaviour can form maximum boiling azeotrope.

➡ Water + Nitric acid shows negative deviation.